

Bio-based Carbon Capture Using Microalgae Systems

Abstract: This research explores the design and modeling of a bio-based carbon capture system using microalgae. The study evaluates CO₂ absorption efficiency, growth kinetics, and system optimization for sustainable carbon sequestration.

Introduction: Climate change driven by CO₂ emissions has led to increasing interest in sustainable carbon capture technologies. Microalgae offer a promising solution due to their high photosynthetic efficiency and rapid growth rates.

Methodology: The system design includes a photobioreactor where algae are cultivated under controlled light, temperature, and CO₂ conditions. Mathematical models are used to simulate CO₂ uptake and biomass growth.

Mathematical Model: The growth rate of algae is modeled using Monod kinetics, while CO₂ absorption is described using mass transfer equations.

Results and Discussion: Simulations indicate that optimized light intensity and CO₂ concentration significantly improve biomass productivity and carbon capture efficiency.

Conclusion: Algae-based carbon capture systems present a viable and scalable solution for reducing atmospheric CO₂. Further work includes experimental validation and scale-up design.

Future Work: Integration with industrial flue gas systems and economic feasibility analysis.